

Lesson 4.1 Balanced Incomplete Block Design

Introduction

Blocking (or stratification or pairing) is a fundamental aspect of statistics. The idea behind blocking is to control for sources of variation that make it difficult to detect differences in the mean response between treatments. For example, in the stream slope example, the density of fish will vary among streams due to intrinsic differences among streams (e.g. different productivity) that is difficult to measure.

Consequently, a good experiment will examine all treatments (e.g. the different stream slopes) in all streams so that the intrinsic differences in density among streams will “cancel out” when comparisons among treatments are made.

In Stat 141, blocks were complete in that every block had every treatment occurring at least once and every treatment occurred in every block at least once. In some cases, blocks are incomplete, either by design (e.g. blocks are too small) or by accident (e.g. missing values).

Below are situations where incomplete block designs are necessary:

- Blocks cannot accommodate the complete set of treatments
- Difficult to find a large set of homogeneous experimental units
- Shortage of animals in a pen
- Shortage of experimental apparatus
- Large number of treatments

In incomplete block designs treatments are arranged in groups that are smaller than a complete replication to eliminate the effect of heterogeneity.

Some care needs to be taken in the analysis of incomplete block designs, but a basic analysis is straightforward with modern software.

An incomplete block design may be balanced or partially balanced. We shall discuss first Balanced Incomplete Block Designs (BIBD).

Balanced Incomplete Block Design

A balanced incomplete block design (BIBD) is an incomplete block design in which all pairs of treatments occur together within a block an equal number of times.

Suppose there are t treatments and b blocks and each block can accommodate exactly $k < t$ treatments only. Further, suppose that the treatments occur r (replicates) times in the entire experiment. Then, the number of times any pair of treatments occur within a block is

$$\lambda = \frac{r(k-1)}{t-1}$$

For example, suppose a chemical engineer suspects that time of reaction for a chemical process is affected by the type of catalyst used. Four catalysts (Cat1, Cat2, Cat3, Cat4) are investigated. The experimental procedure consists of selecting a batch of raw material, loading the pilot plant, and applying each catalyst in separate runs and observing the reaction time. Since variations in the batches of raw materials may affect the chemical process, the engineer decides to use batches as blocks. However, each batch is only large enough to permit three catalysts to be run. The design is given below.

Batch 1	Batch 2	Batch 3	Batch 4
Cat1	Cat1	Cat2	Cat1
Cat3	Cat2	Cat3	Cat2
Cat4	Cat3	Cat4	Cat4

In this case we have

$$\lambda = \frac{r(k-1)}{t-1} = \frac{3(3-1)}{4-1} = 2$$

It can be seen that the pair Cat1 and Cat3 occur in Batch 1 and Batch 2; Cat1 and Cat4 can be found in Batch 1 and Batch 4, and so on.

Linear Model

$$Y_{ij} = \mu + \tau_i + \beta_j + \epsilon_{ij}, \quad i = 1, 2, \dots, t; \quad j = 1, 2, \dots, b$$

where:

- Y_{ij} is the observation for the i^{th} treatment in the j^{th} block
- μ is the overall mean
- τ_i is the effect of the i^{th} treatment

- β_j is the effect of the j^{th} block
- ϵ_{ij} is the random error component, $\epsilon_{ij} \sim N(0, \sigma^2)$

Sums of Squares

The total sum of squares (SST) is calculated as usual.

$$SST = \sum_{i=1}^t \sum_{j=1}^b y_{ij}^2 - \frac{y_{..}^2}{rt}$$

The (unadjusted) block sum of squares is calculated as

$$SSB = \frac{1}{k} \sum_{j=1}^b y_{.j}^2 - \frac{y_{..}^2}{rt}$$

While, the (adjusted) sum of squares for treatments is computed as

$$SSTr_{(adj)} = \frac{k}{\lambda t} \sum_{i=1}^t Q_i^2$$

where Q_i stands for the adjusted total of the i^{th} treatment and is given by

$$Q_i = y_{i.} - \frac{1}{k} \sum_{j=1}^b n_{ij} y_{.j}, \quad n_{ij} = \begin{cases} 1, & \text{if treatment } i \text{ appears in block } j \\ 0, & \text{otherwise} \end{cases}$$

Finally,

$$SSE = SST - SSTr_{(adj)} - SSB$$

Suppose the following data was recorded from the chemical process experiment.

Catalyst	Batches				y _{i.}
	1	2	3	4	
1	73	74	-	71	218
2	-	75	67	72	214
3	73	75	68	-	216
4	75	-	72	75	222
y _{.j}	221	224	207	218	870

It can be verified that

$$\begin{aligned}Q_1 &= 218 - \frac{1}{3}[1(221) + 1(224) + 0(207) + 1(218)] = -\frac{9}{3} \\Q_2 &= 214 - \frac{1}{3}[0(221) + 1(224) + 1(207) + 1(218)] = -\frac{7}{3} \\Q_3 &= 216 - \frac{1}{3}[1(221) + 1(224) + 1(207) + 0(218)] = -\frac{4}{3} \\Q_4 &= 222 - \frac{1}{3}[1(221) + 0(224) + 1(207) + 1(218)] = \frac{20}{3}\end{aligned}$$

and

$$\begin{aligned}SST &= \sum_{i=1}^t \sum_{j=1}^b y_{ij}^2 - \frac{y_{..}^2}{rt} \\&= 63156 - \frac{870^2}{12} \\&= 81 \\SSTr_{(adj)} &= \frac{3}{2(4)} \left[\left(-\frac{9}{3}\right)^2 + \left(-\frac{7}{3}\right)^2 + \left(-\frac{4}{3}\right)^2 + \left(\frac{20}{3}\right)^2 \right] \\&= 22.75 \\SSB &= \frac{1}{3}[221^2 + 224^2 + 207^2 + 218^2] - \frac{870^2}{12} \\&= 55 \\SSE &= 81 - 22.75 - 55 = 3.25\end{aligned}$$

Hence, the ANOVA table is

SoV	df	SS	MS	F	p
Catalyst	3	22.75	7.58	11.66	0.011
Batch	3	55.00			
Error	5	3.25	0.65		
Total	11	81			

Testing for Block Effect (Optional)

In the analysis that we have just made, the total sum of squares is partitioned as

$$SST = SSTr_{(adj)} + SSB + SSE.$$

This is appropriate if we want to test for treatment effect. But if we want to test for block effect, then we partition the total SS as follows:

$$SST = SSTr + SSB_{(adj)} + SSE$$

where:

$$SSTr = \frac{1}{r} \sum_{i=1}^t y_{i.}^2 - \frac{y_{..}^2}{n}$$

and

$$SSB_{(adj)} = \frac{r}{\lambda b} \sum_{j=1}^b Q_j'^2$$

where

$$Q_j' = y_{.j} - \frac{1}{k} \sum_{i=1}^t n_{ij} y_{i.}, \quad n_{ij} = \begin{cases} 1, & \text{if treatment } i \text{ appears in block } j \\ 0, & \text{otherwise} \end{cases}$$

Based on the above formulas, the revised ANOVA table becomes [VERIFY!]

SoV	df	SS	MS	F	p
Catalyst (Adj)	3	22.75	7.58	11.66	0.011
Catalyst (Unadj)	3	11.67			
Batch (Adj)	3	66.08	22.03	33.90	0.001
Batch (Unadj)	3	55.00			
Error	5	3.25	0.65		
Total	11	81			

Comparison of adjusted treatment means

The adjusted treatment effect and its standard error are given as

$$\hat{\tau}_i = \frac{kQ_i}{\lambda t}$$

$$se(\hat{\tau}_i) = \sqrt{\frac{kMSE}{\lambda t}}$$

In addition, the standard error of the difference between two adjusted treatment effects is

$$se(\hat{\tau}_i - \hat{\tau}_j) = \sqrt{\frac{2kMSE}{\lambda t}}, \quad i \neq j$$

For example, suppose we want to compare adjusted Catalysts 1 and 2:

$$\hat{\tau}_1 = \frac{3(-9/3)}{2(4)} = -\frac{9}{8}$$

$$\hat{\tau}_2 = \frac{3(-7/3)}{2(4)} = -\frac{7}{8}$$

$$\implies |\hat{\tau}_1 - \hat{\tau}_2| = \frac{1}{4}$$

Using LSD test at the 5% level of significance, we have

$$LSD_{0.05} = t_{0.025,5} \sqrt{\frac{2kMSE}{\lambda t}} = 2.571 \sqrt{\frac{2(3)(0.65)}{2(4)}} = 1.975$$

which indicates that the difference between Catalysts 1 and 2 is not significant at the 5% level of significance.

Recovery of Intrablock Information

The analysis employed in the previous example is called **intrablock** analysis since block differences are eliminated and all contrasts in the treatment effects can be expressed as comparisons between observations in the same block.

If $\beta_j \sim (0, \sigma_\beta^2)$ and uncorrelated, then additional information on the treatment effects can be obtained. We call this as **interblock analysis** (Yates, 1940).

Consider the block totals

$$y_{.j} = k\mu + \sum_{i=1}^t n_{ij}\tau_i + \underbrace{\left(k\beta_j + \sum_{i=1}^t \epsilon_{ij} \right)}_{\text{error term}}$$

By using the method of least squares and using the constraint

$$\sum_{i=1}^t \hat{\tau}_i = 0$$

we obtain the following interblock estimators of μ and τ_i

$$\begin{aligned}\tilde{\mu} &= \bar{y}_{..} \\ \tilde{\tau}_i &= \frac{\sum_{j=1}^b n_{ij}y_{.j} - kr\bar{y}_{..}}{r - \lambda}\end{aligned}$$

The interblock estimators of τ_i for the previous example are:

$$\begin{aligned}\tilde{\tau}_1 &= \frac{663 - 3(3)(72.5)}{3 - 2} = 10.50 \\ \tilde{\tau}_2 &= \frac{649 - 3(3)(72.5)}{3 - 2} = -3.50 \\ \tilde{\tau}_3 &= \frac{652 - 3(3)(72.5)}{3 - 2} = -0.50 \\ \tilde{\tau}_4 &= \frac{646 - 3(3)(72.5)}{3 - 2} = -6.50\end{aligned}$$

By combining the intrablock and interblock estimators we obtain a single, unbiased, minimum variance estimator of τ_i . Such estimator is a weighted linear combination of the interblock and intrablock estimators. The estimator takes the form

$$\tau_i^* = \alpha_1 \hat{\tau}_i + \alpha_2 \tilde{\tau}_i$$

where the weights are

$$\begin{aligned}\alpha_1 &= \frac{u_1}{u_1 + u_2} \\ \alpha_2 &= \frac{u_2}{u_1 + u_2}, \\ u_1 &= \frac{1}{V(\hat{\tau}_i)} \\ u_2 &= \frac{1}{V(\tilde{\tau}_i)},\end{aligned}$$

and

$$\begin{aligned}V(\hat{\tau}_i) &= \frac{k(t-1)}{\lambda t} \sigma^2 \\ V(\tilde{\tau}_i) &= \frac{k(t-1)}{t(r-\lambda)} (\sigma^2 + k\sigma_\beta^2)\end{aligned}$$

The variances σ^2 and σ_β^2 are estimated by

$$\hat{\sigma}^2 = MSE$$

$$\hat{\sigma}_\beta^2 = \begin{cases} \frac{(b-1)[MSB_{Adj} - MSE]}{t(r-1)}, & \text{if } MSB_{Adj} > MSE \\ 0, & \text{otherwise} \end{cases}$$

Hence, the combined estimator takes this final form:

$$\tau_i^* = \begin{cases} \frac{kQ_i(\hat{\sigma}^2 + k\hat{\sigma}_\beta^2) + \left(\sum_{j=1}^b n_{ij}y_{.j} - kr\bar{y}_{..} \right) \hat{\sigma}^2}{(r-\lambda)\hat{\sigma}^2 + \lambda t(\hat{\sigma}^2 + k\hat{\sigma}_\beta^2)}, & \text{if } \hat{\sigma}_\beta^2 > 0 \\ \frac{y_{i.} - \frac{1}{t}y_{..}}{r}, & \text{if } \hat{\sigma}_\beta^2 = 0 \end{cases}$$

For the same example,

$$\begin{aligned} \hat{\sigma}^2 &= MSE = 0.65 \\ \hat{\sigma}_\beta^2 &= \frac{(b-1)[MSB_{Adj} - MSE]}{t(r-1)} \\ &= \frac{(4-1)[22.03 - 0.65]}{4(3-1)} \\ &= 8.02 \end{aligned}$$

Thus, the combined estimator of the effect of Catalyst 1 is calculated as follows:

$$\begin{aligned} Q_1 &= -3, \text{ from earlier computations} \\ \sum_{j=1}^b n_{ij}y_{.j} - kr\bar{y}_{..} &= (221 + 224 + 218) - 3(3)(72.5) = 10.5 \\ \tau_1^* &= \frac{3(-3)[0.65 + 3(8.02)] + 10.5(0.65)}{(3-2)0.65 + 2(4)[0.65 + 3(8.02)]} = -1.09 \end{aligned}$$

The rest of the intrablock, interblock, and combined effect estimates are shown in the table below. {VERIFY!}

Parameter	Intrablock	Interblock	Combined
τ_1	-1.12	10.50	-1.09
τ_2	-0.88	-3.50	-0.88
τ_3	-0.50	-0.50	-0.50
τ_4	2.50	-6.50	2.47

Analysis of BIBDs using R

There are many functions in R that can be used to analyze BIBDs. First, we load the required packages and the data set.

```
#Loading required packages and the data set
library(agricolae)
library(car)
library(tidyverse)
library(readxl)
library(emmeans)
library(ibd)

bibd_data <- read_excel("BIBD.data.xlsx")

bibd_data <- bibd_data %>%
  mutate(Catalyst = factor(Catalyst),
         Block = factor(Block))

head(bibd_data)
```

```
## # A tibble: 6 x 3
##   Catalyst Block  Time
##   <fct>    <fct> <dbl>
## 1 1      1      73
## 2 1      2      74
## 3 1      4      71
## 4 2      2      75
## 5 2      3      67
## 6 2      4      72
```

We can use the built-in *aov()* function to run the analysis and the *anova()* function to display ANOVA table containing the Type I SS. If we want to generate the adjusted SS for the treatments, the block term must be added first in the model, and vice-versa. These are illustrated in the following code chunks.

```
#Will calculate the adjusted catalyst SS and the unadjusted block SS
out1 <- aov(Time ~ Block + Catalyst,
            data = bibd_data)
```

```
anova(out1)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Time
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Block      3  55.00  18.3333   28.205 0.001468 **
## Catalyst   3   22.75   7.5833   11.667 0.010739 *
## Residuals  5    3.25   0.6500
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#Will calculate the adjusted block SS and the unadjusted catalyst SS
out2 <- aov(Time ~ Catalyst + Block,
            data = bibd_data)
```

```
anova(out2)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Time
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Catalyst   3  11.667   3.8889   5.9829 0.0414634 *
## Block      3  66.083  22.0278  33.8889 0.0009528 ***
## Residuals  5   3.250   0.6500
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Alternatively, we can request for a Type II SS using the *Anova()* function in the **car** package.

```
#Will calculate both the adjusted catalyst and block SS
out3 <- aov(Time ~ Catalyst + Block,
            data = bibd_data)

Anova(out3, type = 2)
```

```
## Anova Table (Type II tests)
##
## Response: Time
##           Sum Sq Df F value    Pr(>F)
## Catalyst  22.750  3  11.667 0.0107387 *
## Block     66.083  3  33.889 0.0009528 ***
## Residuals  3.250  5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

To compute the adjusted Catalyst (treatment) means we use the *emmeans()* function in the **emmeans** package.

```
emmeans(out1, "Catalyst")
```

```
## Catalyst emmean    SE df lower.CL upper.CL
## 1           71.4 0.487  5     70.1     72.6
## 2           71.6 0.487  5     70.4     72.9
## 3           72.0 0.487  5     70.7     73.3
## 4           75.0 0.487  5     73.7     76.3
##
## Results are averaged over the levels of: Block
## Confidence level used: 0.95
```

There is also the **agricolae** package which contains functions for the generation and analysis of IBIDs. To illustrate, let us generate a BIBD with $t = 4$, $b = 4$, $k = 3$, and $r = 3$.

```
trt <- c("A", "B", "C", "D")
outdesign <- design.bib(trt, k = 3, seed = 1234)
```

```
##
## Parameters BIB
## =====
## Lambda      : 2
## treatmeans  : 4
## Block size  : 3
## Blocks      : 4
## Replication: 3
##
## Efficiency factor 0.8888889
##
## <<< Book >>>
```

```
print(outdesign$sketch)
```

```
##      [,1] [,2] [,3]
## [1,] "B"  "A"  "C"
## [2,] "B"  "D"  "A"
## [3,] "A"  "C"  "D"
## [4,] "C"  "B"  "D"
```

From the *sketch* we run treatments B, A, and C in Block 1; B, D, and A in Block 2; A, C, and D in Block 3; and C, B, and D in Block 4.

To analyze the data in the current example, we have the following chunk.

```
BIB.test(block = bibd_data$Block,
          trt = bibd_data$Catalyst,
          y = bibd_data$Time,
          test="tukey",
          console=TRUE,
          group = TRUE)
```

```
##
## ANALYSIS BIB:  bibd_data$Time
## Class level information
##
```

```
## Block:  1 2 4 3
## Trt   :  1 2 3 4
##
## Number of observations:  12
##
## Analysis of Variance Table
##
## Response: bibd_data$Time
##           Df Sum Sq Mean Sq F value    Pr(>F)
## block.unadj  3  55.00  18.3333   28.205 0.001468 **
## trt.adj      3   22.75   7.5833   11.667 0.010739 *
## Residuals    5    3.25   0.6500
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## coefficient of variation: 1.1 %
## bibd_data$Time Means: 72.5
##
## bibd_data$Catalyst,  statistics
##
## bibd_data$Time mean.adj      SE r      std Min Max
## 1      72.66667    71.375 0.4868051 3 1.527525  71  74
## 2      71.33333    71.625 0.4868051 3 4.041452  67  75
## 3      72.00000    72.000 0.4868051 3 3.605551  68  75
## 4      74.00000    75.000 0.4868051 3 1.732051  72  75
##
## Tukey
## Alpha      : 0.05
## Std.err    : 0.4937104
## HSD        : 2.576341
## Parameters BIB
## Lambda     : 2
## treatmeans : 4
## Block size : 3
## Blocks     : 4
## Replication: 3
##
## Efficiency factor 0.8888889
##
## <<< Book >>>
##
## Comparison between treatments means
##           Difference pvalue sig.
## 1 - 2      -0.250 0.9825
## 1 - 3      -0.625 0.8085
```

```
## 1 - 4      -3.625 0.0130      *
## 2 - 3      -0.375 0.9462
## 2 - 4      -3.375 0.0175      *
## 3 - 4      -3.000 0.0281      *
##
## Treatments with the same letter are not significantly different.
##
## bibd_data$Time groups
## 4          75.000      a
## 3          72.000      b
## 2          71.625      b
## 1          71.375      b
```

Finally, there is the **ibd** package which contains many useful functions for the generation and analysis of IBDs. For example, the *aov.ibd()* function does similar thing to what the *Anova()* function does.

```
options(digits = 10)
out4 <- aov.ibd(Time ~ Catalyst + Block,
               data = bibd_data)
out4
```

```
## $ANOVA.table
## Anova Table (Type III tests)
##
## Response: Time
##              Sum Sq Df    F value    Pr(>F)
## (Intercept) 8948.6786  1 13767.19780 8.5281e-10 ***
## Catalyst      22.7500  3   11.66667 0.01073866 *
## Block         66.0833  3   33.88889 0.00095276 ***
## Residuals      3.2500  5
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```